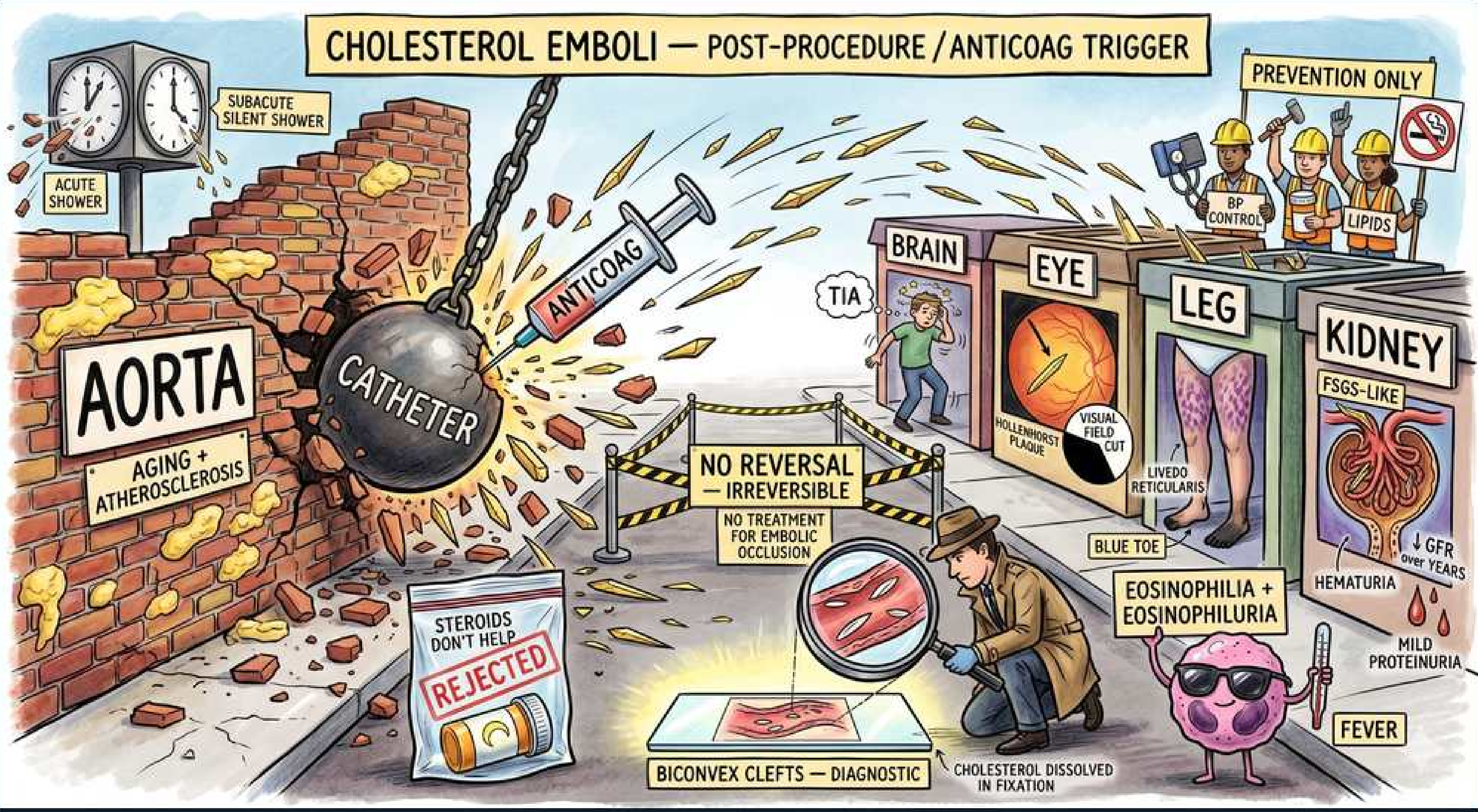
Glomerular (Nephrotic) — Cholesterol (Atheroembolic) Nephropathy



1. Post-procedure / anticoag trigger

WHAT YOU SEE

Top banner CHOLESTEROL EMBOLI — POST-PROCEDURE / ANTICOAG TRIGGER

WHAT IT ENCODES

Cholesterol (atheroembolic) disease is a showering of needle-shaped cholesterol crystals from ulcerated aortic plaques into small arteries (150–200 μm), where they lodge and incite a foreign-body inflammatory reaction with intimal proliferation and ischemia. Three classic triggers: (1) intra-arterial instrumentation — cardiac catheterization, coronary/peripheral angiography, aortic surgery or stenting; (2) anticoagulation (heparin, warfarin); (3) thrombolysis (tPA). The unifying patient is an older man with diffuse atherosclerosis, a smoking history, and recent vascular intervention.

BOARD TRAP

On boards, the wrong answer is 'contrast-induced nephropathy.' Discriminator: contrast nephropathy is a direct tubular/hemodynamic insult peaking at 24–72h with FULL recovery in ~7 days, normal complement, no eosinophilia, and no skin findings. Atheroembolism is a mechanical/inflammatory disease with subacute AKI over 1–6 WEEKS, eosinophilia, low complement, livedo, and blue toes — and it is provoked by manipulation/anticoagulation, not by the dye itself.

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2. Aorta / aging atherosclerosis

WHAT YOU SEE

Wall labeled AORTA — aging + atherosclerosis

WHAT IT ENCODES

The embolic source is an ulcerated, friable atherosclerotic plaque in the aorta or other large vessel. Risk scales with age, diffuse atherosclerosis, prior MI/stroke/PAD, hypertension, smoking, and high LDL. The aorta is the 'crumbling wall' that sheds crystals downstream to kidney, skin, gut, retina, and brain — explaining the multi-organ picture.

BOARD TRAP

Wrong answer: 'large-vessel thromboembolism from atrial fibrillation.' Discriminator — AFib emboli are platelet/fibrin thrombi that occlude a single large vessel acutely (e.g., a cold pulseless limb or a wedge-shaped renal infarct), whereas cholesterol crystals are tiny, shower diffusely, and leave distal pulses intact.

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3. Catheter wrecking ball

WHAT YOU SEE

Wrecking ball labeled CATHETER smashing aortic wall

WHAT IT ENCODES

The catheter is the wrecking ball: mechanical scraping of the aortic intima during angiography or PCI physically dislodges cholesterol crystals from the plaque core. Incidence of clinically apparent disease after cardiac cath is roughly 1–2% but is far higher (>25%) on autopsy series, so most cases are subclinical. The more aortic manipulation, the higher the embolic load.

BOARD TRAP

Wrong answer: 'the catheter caused acute tubular necrosis from contrast.' Discriminator — the injury here is mechanical plaque disruption (a crystal embolus), not osmotic/ischemic tubular damage. The tell is the timeline (weeks, not 1–3 days) plus systemic embolic stigmata.

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4. Anticoag syringe trigger

WHAT YOU SEE

Syringe labeled ANTICOAG aimed at plaque

WHAT IT ENCODES

Anticoagulants and thrombolytics destabilize plaque by lysing or preventing the protective fibrin cap that overlies the ulcerated lesion, exposing the cholesterol core to the bloodstream and unleashing crystal showers. This is why a patient started on warfarin/heparin or given tPA can develop atheroembolism days to weeks later.

BOARD TRAP

Wrong answer: 'start/continue anticoagulation to treat the emboli.' Discriminator — anticoagulation is a CAUSE and can WORSEN atheroembolism; there is no role for it (and you generally hold it). Contrast this with true thromboembolism (e.g., renal vein thrombosis or cardioembolic infarct) where anticoagulation IS the treatment.

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5. Subacute silent shower

WHAT YOU SEE

Top-left clocks 'subacute silent shower' / 'acute shower'

WHAT IT ENCODES

The clocks emphasize TIMING. Renal failure from atheroembolism is typically subacute and stepwise over 1–6 weeks, reflecting repeated crystal showers rather than a single event; a smaller subset has an acute shower with rapid decline. Creatinine often rises in a saltatory, stuttering pattern and may never fully recover.

BOARD TRAP

Wrong answer: 'AKI within 24–72h of the procedure = atheroembolism.' Discriminator — that fast onset points to contrast nephropathy. Atheroembolic AKI is delayed (days to weeks) and progressive, which is the single most-tested timing distinction on the exam.

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6. Blue toe / livedo reticularis

WHAT YOU SEE

Leg panel 'livedo reticularis' and 'BLUE TOE'

WHAT IT ENCODES

Cutaneous signs are the highest-yield clue: 'blue toe syndrome' (painful, cyanotic, well-demarcated toes) and livedo reticularis (lacy, reticular violaceous mottling, usually on the legs/trunk). Other findings: digital gangrene, Hollenhorst plaques, and splinter-like lesions. Skin findings appear in roughly a third to half of cases.

BOARD TRAP

Wrong answer: 'acute limb ischemia from arterial occlusion — pulses should be absent.' Discriminator — in atheroembolism the DISTAL PULSES ARE PRESERVED because the occluded vessels are tiny (downstream of palpable arteries). Blue toes with bounding pulses is the classic mismatch that points away from large-vessel disease.

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7. Eye / Hollenhorst plaque

WHAT YOU SEE

Eye panel 'visual field' (retinal cholesterol plaque)

WHAT IT ENCODES

Fundoscopy can show a Hollenhorst plaque — a bright, refractile, yellow-orange cholesterol crystal lodged at a retinal arteriole bifurcation, sometimes with a visual field defect or amaurosis fugax. It is direct visual proof of an aortic/carotid embolic source and supports the diagnosis non-invasively.

BOARD TRAP

Wrong answer: 'a Hollenhorst plaque indicates retinal vein occlusion.' Discriminator — Hollenhorst plaques are ARTERIAL cholesterol emboli; retinal vein occlusion shows tortuous dilated veins and a 'blood-and-thunder' fundus, a completely different mechanism.

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8. Brain / TIA

WHAT YOU SEE

A small brain panel labeled TIA among the row of organ panels (brain/eye/leg/kidney) — it shows crystals showering the brain too, making this a diffuse multi-organ embolic syndrome rather than a single-organ event.

WHAT IT ENCODES

CNS crystal showers can produce TIA, stroke, or transient confusion as part of the multi-organ embolic syndrome. Together with gut involvement (mesenteric ischemia, GI bleeding, abdominal pain) and renal failure, this multi-system picture is what distinguishes a diffuse embolic shower from an isolated organ event.

BOARD TRAP

Wrong answer: 'multi-organ involvement plus low complement and eosinophilia = systemic vasculitis (e.g., ANCA or polyarteritis nodosa).' Discriminator — atheroembolism mimics vasculitis perfectly but has a clear procedural/anticoagulation trigger, NORMAL ANCA, and biconvex cholesterol clefts on biopsy rather than fibrinoid necrosis.

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9. Eosinophilia + eosinophiluria

WHAT YOU SEE

Detective magnifier 'EOSINOPHILIA + EOSINOPHILURIA'

WHAT IT ENCODES

Laboratory signature: peripheral eosinophilia (transient, ~60–80% of cases), eosinophiluria, transient HYPOCOMPLEMENTEMIA (low C3/C4), elevated ESR/CRP, and sometimes fever and weight loss. These reflect complement activation and a systemic inflammatory response to the crystals. The combination is a major diagnostic clue when paired with skin findings and recent instrumentation.

BOARD TRAP

Wrong answer: 'eosinophilia + AKI = allergic interstitial nephritis (AIN), give steroids.' Discriminator — AIN follows a culprit DRUG (PPIs, NSAIDs, beta-lactams) with WBC casts/sterile pyuria, not a procedure; it lacks livedo, blue toes, and low complement. Atheroembolism is the answer when the eosinophilia comes after a cath plus skin stigmata.

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10. Biconvex clefts diagnostic

WHAT YOU SEE

Tissue under magnifier 'BICONVEX CLEFTS — diagnostic'

WHAT IT ENCODES

The definitive (though often unnecessary) test is biopsy of an affected organ (skin, kidney, muscle) showing needle-shaped, biconvex CHOLESTEROL CLEFTS occluding small arteries/arterioles, surrounded by a foreign-body giant-cell reaction. The clefts appear as empty slit-like spaces because the cholesterol itself dissolves out during alcohol/xylene tissue processing.

BOARD TRAP

Wrong answer: 'the empty clefts mean the slide is artifact or the lesion is non-specific.' Discriminator — the empty biconvex spaces ARE the pathognomonic finding; cholesterol is lipid-soluble and washes out in fixation, leaving the characteristic ghost shape. Diagnosis is usually clinical, so biopsy is reserved for unclear cases.

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11. No reversal / irreversible

WHAT YOU SEE

Center barricade 'NO REVERSAL — IRREVERSIBLE / no treatment for embolic occlusion'

WHAT IT ENCODES

There is NO therapy that dissolves lodged crystals or reverses established embolic occlusion. Management is purely supportive: aggressive statin therapy (e.g., atorvastatin 40–80 mg or rosuvastatin 20–40 mg, plaque stabilization), strict BP control, avoid further arterial instrumentation, and stop/avoid anticoagulation where possible. Dialysis if needed; prognosis is guarded with significant mortality.

BOARD TRAP

Wrong answer: 'urgent thrombolysis, embolectomy, or therapeutic anticoagulation will restore perfusion.' Discriminator — these target THROMBI; crystals cannot be lysed or removed and anticoagulation/thrombolysis are themselves precipitants. The correct move is supportive care plus statin and risk-factor control.

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12. Steroids don't help / rejected

WHAT YOU SEE

Bottom-left bag 'STEROIDS DON'T HELP — REJECTED'

WHAT IT ENCODES

Despite eosinophilia, low complement, and an inflammatory appearance that screams 'steroid-responsive,' corticosteroids are NOT proven to change outcomes in atheroembolic disease and are not standard therapy. The eosinophilia is a clue to the diagnosis, not an indication for immunosuppression.

BOARD TRAP

Wrong answer: 'give high-dose corticosteroids for the eosinophilia/low complement.' Discriminator — steroids treat AIN and true vasculitis, both of which atheroembolism mimics. The presence of a procedural trigger + blue toes + cholesterol clefts tells you steroids are the wrong (and ineffective) choice.

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13. Prevention only / risk control

WHAT YOU SEE

Top-right workers 'PREVENTION ONLY — BP control / lipids'

WHAT IT ENCODES

Because the disease is irreversible, the only truly effective strategy is PREVENTION: high-intensity statins to stabilize plaque, tight blood-pressure and lipid control, smoking cessation, and minimizing unnecessary arterial catheterization and anticoagulation in high-risk atherosclerotic patients. Radial (vs femoral) access reduces embolic risk during coronary procedures.

BOARD TRAP

Wrong answer: 'aggressive procedural intervention to clear the aorta.' Discriminator — more instrumentation = more crystal showers. The exam reward is risk-factor modification and AVOIDING further arterial manipulation, not a procedural fix.

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14. Kidney FSGS-like / hematuria

WHAT YOU SEE

Right KIDNEY panel 'FSGS-like / ↓GFR over years / hematuria / mild proteinuria'

WHAT IT ENCODES

The renal picture is rising creatinine with SUB-nephrotic (mild) proteinuria, occasional hematuria, and a stepwise/saltatory GFR decline; chronic glomerular changes can resemble focal segmental glomerulosclerosis (FSGS). Renal recovery is often incomplete and many patients progress to CKD or dialysis.

BOARD TRAP

Wrong answer: 'expect heavy nephrotic-range proteinuria (>3.5 g/day) with edema.' Discriminator — atheroembolic renal disease is typically a vascular/ischemic process giving MILD proteinuria; full nephrotic syndrome points to a primary glomerular disease (FSGS, membranous, diabetic, amyloid) instead.
